

Semantic Identity Compression: Exact Zero-Error Laws, Rate-Distortion, and Neurosymbolic Necessity

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Abstract

The question of whether semantic representations can capture identity without auxiliary identifiers is often framed as a design question. This paper shows that it is governed by exact coding constraints. When distinct entities map to the same representation, no decoder can recover their identities without additional distinguishing information. The minimum worst-case auxiliary description is exactly $\log_2 A_\pi$ bits, where A_π is the maximum collision multiplicity. If this cost is left unpaid, the distortion floor on a worst collision block is $D \geq 1 - 1/A_\pi$.

The paper isolates the deterministic corner of this problem. On a uniform collision block of size a with auxiliary budget L bits, the optimal distortion is exactly $D^*(L) = \max(0, 1 - 2^L/a)$. For repeated independent uses of the same representation map, the minimum zero-error block budget is $L_t^* = t \log_2 A_\pi$. In the adaptive regime, the optimal fiber budget is $\lceil \log_2 |\pi^{-1}(u)| \rceil$, any feasible adaptive scheme has expected length at least the expectation of that quantity and hence at least $H(C | U)/\log 2$, and standard fiberwise prefix coding achieves expected length at most $H(C | U)/\log 2 + 1$. The point is that a question often discussed at the level of architecture can be stated and answered as an exact coding problem.

The same ambiguity-fiber geometry yields exact task-sufficiency, helper-view, and factorized-representation consequences. We instantiate these generic laws in a concrete formal identity domain and show that a neural encoder paired with a symbolic handle achieves zero-error identity recovery if and only if the handle resolves ambiguity inside each semantic fiber ([NSL1–NSL6](#)). Auxiliary distinguishing

information must be supplied somewhere; symbolic layers are a standard computational mechanism for supplying it.

Keywords: semantics-aware compression, zero-error coding, neurosymbolic systems, learned representations, side information

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I. INTRODUCTION

A. Problem Statement

Learned semantic representations, embeddings, concept bottlenecks, and neural encoders compress entities into finite intermediate states. A basic question is whether those representations alone can support zero-error identity recovery in an open world.

This paper proves the answer is no whenever the representation has collisions. If two distinct entities map to the same semantic representation, then no decoder can recover their identities without additional information. The required distinguishing information must therefore be supplied either by an injective encoder or by an auxiliary identity mechanism such as a symbolic handle.

Formally, let C be a class label and let $U = \pi(C)$ be a finite representation available to the decoder. When π is noninjective, the representation alone does not determine the class. The question in this paper is: how much additional description must be transmitted to recover C exactly?

This is a zero-error compression problem with side information at the decoder. The side information is the representation value U , and the encoder sends auxiliary description M only to resolve collisions inside the fiber $\pi^{-1}(U)$. The paper studies this problem for any fixed representation map π , whether it arises from learned embeddings, schema projections, feature extraction pipelines, or an explicit attribute family. It shows that the same coding object induces a deterministic semantic identity rate-distortion curve when the budget is subcritical or zero, and then compares those transmission costs with two complementary modalities: query costs for operational disambiguation and robustness costs under world growth.

The paper should be read in two connected arcs. First, it gives exact coding laws for the residual ambiguity left by a fixed representation. Second, it shows that the same ambiguity-fiber object yields exact representation-sufficiency laws for downstream tasks, helper views, and modular representations.

The basic combinatorial step is simple: once the decoder already sees the representation, exact recovery reduces to resolving the residual ambiguity inside its fibers. The first arc is the exact finite backbone of the theory; the broader payoff is the wider yield of that same primitive across fixed-length, adaptive, entropy, distortion, and downstream representation-sufficiency consequences.

All theorems are relative to one explicit admissibility contract: the decoder may use only the declared representation value and any auxiliary description whose length is charged to the coding budget. No hidden registry, preprocessing table, or external side channel may contribute uncharged distinguishing information.

B. Why This Is a Compression Problem

The central object is the fiber geometry induced by π . Let

$$A_\pi := \max_u |\{c : \pi(c) = u\}|$$

be the largest collision fiber. Then the fixed-length zero-error question is exactly the minimum number of extra bits needed to separate the worst fiber. The answer is an exact lossless coding law:

$$L \geq \log_2 A_\pi,$$

and this lower bound is tight. Thus the representation itself acts as decoder side information, and the remaining coding burden is precisely the ambiguity left unresolved by the representation.

This viewpoint also yields a finer representation-adaptive question. If the encoder may vary the extra description length with the observed representation value u , then the optimal pointwise budget is governed by the size of the individual fiber $\pi^{-1}(u)$. The resulting expected-length theorem is the average-case counterpart of the worst-case max-fiber law. It also yields a sharp

rate-distortion style statement in the deterministic exact-recovery corner: if one refuses to pay the necessary side-information budget, the distortion is not merely nonzero but bounded below explicitly by the collision geometry. In this sense, worst-case, adaptive, entropy-comparison, and below-threshold distortion statements are all different views of the same residual ambiguity object.

C. Relation to Prior Work

The closest classical objects are zero-error coding with confusability constraints [1], [2], source coding with side information, and more recent task-oriented or semantics-aware communication models [3]. Our setting is deterministic and zero-error on the main theorem path: the representation is fixed, the decoder already observes $U = \pi(C)$, and the coding problem is to quantify the additional description required for exact recovery of C .

The paper does not claim a full graph-entropy characterization, a full noisy rate-distortion region, or a general noisy semantic communication theory. Instead, it isolates an exact zero-error residual-coding problem induced by a fixed representation, solves it exactly in the fixed-length and pointwise fiberwise senses, derives a sharp distortion floor below the exact-recovery threshold, places the adaptive regime between matching lower and standard prefix-style upper bounds, and uses the same fiber geometry to organize several semantic identification settings under one side-information viewpoint.

For the fixed-length zero-error theorem, the confusability graph is a special case: classes sharing one representation value form a clique, so the graph is a disjoint union of representation-fiber cliques. In that setting, the classical zero-error coloring law collapses to the statement that the required alphabet size is the largest fiber size. The interest here is therefore not a new graph-theoretic identity in isolation, but the identification of the correct residual coding object and the consequences that follow from it: exact fixed-length laws, fiberwise adaptive budgets, entropy comparisons, explicit distortion floors, and representation-level interpretations from the same fiber geometry.

The relation to Slepian-Wolf coding is similarly directional rather than identical. Slepian-Wolf studies asymptotically optimal coding with correlated side information, whereas here the

side information is the deterministic fixed representation $U = \pi(C)$ and the emphasis is on exact finite-block zero-error recovery. The common theme is decoder side information; the difference is that the present paper treats the representation map as fixed and asks for the residual coding burden it leaves behind. On the lossy side-information axis, Wyner-Ziv problems play the analogous role: the present paper isolates the deterministic exact-recovery corner before any noisy or rate-distortion refinement.

The modern interpretation is neurosymbolic necessity. If π is interpreted as a learned embedding, then collision fibers measure the irreducible identity ambiguity left by semantic compression. Injective representations need no auxiliary metadata. Noninjective representations impose an exact-recovery cost that no downstream reasoning layer can remove on its own. The concrete formal instantiation developed later in the paper illustrates that claim, not a limitation on it: the zero-error necessity statement applies to any fixed learned representation with collisions. This is especially natural in concept-bottleneck and neuro-symbolic settings, where a fixed intermediate representation must support exact downstream symbolic or class-level decisions [4], [5].

D. Why the Formulation Matters

The core counting fact is simple once the problem is expressed in terms of representation fibers. If A_π classes share the same representation value, then distinguishing them exactly requires at least $\log_2 A_\pi$ bits. The main contribution of the paper is therefore not a technically elaborate converse, but the observation that several questions about semantics, identity, and auxiliary information reduce to this same coding object.

That reduction is easy to miss in practice. Rich semantic representations often seem as though they should capture identity as a special case, but any collision fiber shows otherwise: once distinct entities share a representation value, no decoder can separate them from the representation alone. Closed-world deployments can also hide the issue, since injectivity can sometimes be engineered into the chosen universe. The coding laws matter most when the world is open or the representation is fixed and not built specifically for exact identity.

Error tolerance can further soften the appearance of the problem. Many applications accept some misidentification, so the failure mode may seem gradual. The deterministic-corner distortion theorem shows that the penalty is instead quantitatively sharp. Likewise, systems that appear to rely only on representations often carry auxiliary identifiers in the background, such as retrieval keys, document IDs, memory addresses, cache keys, or registry entries. The paper makes that implicit accounting explicit.

E. Contributions

- 1) **Semantic identity rate-distortion curve on collision blocks.** On a collision block of size a under the uniform source, the optimal distortion under an auxiliary budget of L bits is exactly

$$D^*(L) = \max\left(0, 1 - \frac{2^L}{a}\right)$$

(Theorem IV.12). This makes the paper's rate-distortion contribution explicit in the deterministic corner: below the identity threshold, distortion rises according to an exact curve dictated by collision geometry. In the finite multi-fiber setting, the corresponding recoverable source mass decomposes exactly into fiberwise optima and is attained by optimizing independently within each fiber (FRD1, FRD2).

- 2) **Zero-error threshold and semantic identity rate law.** The zero-error threshold $D^*(L) = 0$ is achieved exactly when $L \geq \log_2 a$ on a collision block (Corollary IV.13). In the full representation, any exact identification scheme requires at least $\log_2 A_\pi$ extra bits in the worst case (Theorem III.11), and this bound is tight via the exact one-shot feasibility threshold (Theorem III.14). For block length t , the minimum fixed-length budget is exactly $L_t^* = t \log_2 A_\pi$ (Corollary III.16). Zero stored auxiliary bits force the floor $D \geq 1 - 1/A_\pi$ on a worst collision block (Corollary IV.14).
- 3) **Adaptive side information and average-case rates.** If the auxiliary description length may depend on the observed representation value u , then feasibility requires at least $\lceil \log_2 |\pi^{-1}(u)| \rceil$ bits on that fiber. Consequently, any feasible adaptive scheme has expected length at least the expectation of this pointwise optimal budget (Theorem IV.4), hence at

least $H(C \mid U)/\log 2$ under any source law. Under standard prefix-coding achievability within each fiber, there also exists an adaptive zero-error code with expected length at most $H(C \mid U)/\log 2 + 1$ (Section IV). Taken together, these give a mechanized zero-error conditional-entropy sandwich (ZEC1). The pointwise fiber budget is exact; the entropy comparison is a lower/upper bound pair rather than an exact equality claim.

- 4) **Neurosymbolic necessity theorems.** We instantiate the generic helper-view framework in a concrete identity domain, proving that semantic representation alone cannot achieve zero-error identity when collisions exist (NSL2) and that no decoder can recover collapsed identities without additional identity information (NSL3). The neurosymbolic linking theorems establish that the pair consisting of a semantic encoder and an injective symbolic handle achieves zero-error identity recovery, with the exact characterization that identity is recoverable if and only if the handle distinguishes all entities within each semantic fiber (NSL1, NSL4, NSL5). This helper-view pattern is canonical for open-world semantic systems (NSL6). The mechanization connects the information-theoretic helper-view framework (GPH40) to a concrete formal identity model (ACS8, ACS9, EMB1). The instantiated domain illustrates the broader claim about learned representations with collisions. The required distinguishing information must be supplied either in the encoder or in the auxiliary identity channel.
- 5) **Representation sufficiency laws.** In the finite explicit setting, one-symbol zero-error feasibility is equivalent to injectivity, and refining the observation channel cannot increase collision multiplicity (GPH27, GPH28). Coarsening a representation therefore cannot reduce either the worst-case fixed-length burden or the pointwise adaptive burden (Corollary III.9). More generally, for any downstream task label $Y = f(C)$, exact recovery from the representation alone is possible if and only if each representation fiber carries at most one task label (Theorem V.2), and coarsening cannot improve this task-level sufficiency (Corollary V.3). Helper views can only reduce that ambiguity (Theorem V.4, Corollary V.5), and in genuinely factorized product settings the residual ambiguity multiplies exactly across modules (Corollary V.6). These results show that representation collisions are not merely a

heuristic notion of quality but an exact zero-error residual-coding burden for downstream identification.

- 6) **Query and stability costs of residual ambiguity.** Residual ambiguity can be paid for in more than one currency. If one does not transmit the needed side information, then exact recovery incurs interactive/query cost, quantified by the minimum distinguishing number d ; equivalently, one must build a sufficiently rich helper view out of primitive observables. Even when a current snapshot is collision-free, ambiguity may return under world growth, and the robustness section quantifies that stability cost. These theorem families are complementary to the coding arc rather than separate topics.

F. Worked Example

We use one small example throughout the paper. Five classes have profiles

$$A = 000, \quad B = 000, \quad C = 100, \quad D = 110, \quad E = 111.$$

Then A and B form a collision fiber of size two, so the exact worst-case fixed-length budget is one bit. The same example also gives a simple adaptive picture: the fiber over profile 000 needs one auxiliary bit, while the singleton fibers need none. In the learned-representation reading, this is the exact price of one residual collision in the representation.

G. Paper Organization

Section II formalizes the generic representation-side-information framework and the basic information barrier. Sections III and IV develop the main coding theorems: zero-error laws, adaptive budgets, and the deterministic-corner semantic identity rate-distortion result. Section V then presents two instantiations of those laws, beginning with learned representations and then turning to a formal entity-attribute model. Readers primarily interested in learned representations may read directly from the generic theorem arc to Section V. Section VI studies query costs as an operational alternative to transmitting side information, and Section VII studies the stability cost of residual ambiguity under system growth. Section VIII closes with extensions and open directions.

II. MODEL AND BASIC INFORMATION BARRIER

A. The Abstract Coding Framework

The paper studies a finite class space together with a fixed representation map. The decoder observes the representation value, and the encoder may additionally send auxiliary description to resolve whatever ambiguity the representation leaves behind. The coding laws depend only on this representation map and its fibers, not on the mechanism that produced it.

Definition II.1 (Coding model induced by a representation). Let C be a finite class variable, let $U = \pi(C)$ be the representation available to the decoder, and let M be any auxiliary description transmitted by the encoder. The main theorem arc studies the minimum length of M needed for exact recovery of C from (U, M) , in both fixed-length and representation-adaptive regimes.

Definition II.2 (Representation fiber). For a representation value u , define the corresponding fiber by

$$\pi^{-1}(u) = \{c : \pi(c) = u\}.$$

The paper's central object is the geometry of these fibers.

Remark II.3 (Where the representation may come from). No further structure is assumed. The map π may arise from a learned encoder, a database schema projection, a feature-extraction pipeline, or a hand-designed attribute family. The theoremic contribution is generic in the representation map once its fibers are fixed.

Remark II.4 (Entity and random-variable notation). The coding statements are written in terms of a class variable C and representation $U = \pi(C)$. When convenient, we also use an underlying entity variable V together with a class assignment map $C(V)$. Thus $U = \pi(V)$ is the representation of the realized entity, and the induced class label is $C(V)$. The two viewpoints are interchangeable; the coding arc should be read through the pair (C, U) .

Definition II.5 (Admissibility contract). All impossibility and optimality statements in this paper quantify over schemes whose evidence consists only of the declared representation value together

with any explicit auxiliary description charged to the side-information budget. In particular, no hidden registry, reflection oracle, whole-universe preprocessing table, or amortized cross-instance cache may supply uncharged distinguishing information.

B. Representation-Induced Information Barrier

The basic obstruction is immediate: equal representations give the decoder the same side information. The next theorem records that invariant in generic form.

Definition II.6 (Representation-computable property). A property $P : \mathcal{C} \rightarrow Y$ is *representation-computable* if there exists a function $f : \mathcal{U} \rightarrow Y$ such that

$$P(c) = f(\pi(c)) \quad \text{for all } c \in \mathcal{C}.$$

Equivalently, P depends only on the representation value.

Theorem II.7 (Information barrier). *Let $P : \mathcal{C} \rightarrow Y$ be any function. If P is representation-computable, then P is constant on every representation fiber:*

$$\pi(c_1) = \pi(c_2) \implies P(c_1) = P(c_2).$$

Equivalently, no observer restricted to the representation alone can compute a property that varies within one fiber.

Proof. Suppose P is representation-computable via f , so $P(c) = f(\pi(c))$ for all c . If $\pi(c_1) = \pi(c_2)$, then

$$P(c_1) = f(\pi(c_1)) = f(\pi(c_2)) = P(c_2).$$

■

Remark II.8 (Role of the barrier). The barrier is informational rather than computational: once the representation is identical, unlimited time or memory does not help. We state it explicitly because the later converse, adaptive coding, query, and robustness results all reuse the same invariance premise.

Corollary II.9 (Class identity is not representation-computable). *If there exist classes c_1, c_2 with $\pi(c_1) = \pi(c_2)$ but $c_1 \neq c_2$, then class identity is not representation-computable.*

Proof. Direct application of Theorem II.7 to the identity map on classes. ■

Theorem II.10 (Observable-state sufficiency). *Let $\alpha : \mathcal{C} \rightarrow \mathcal{A}$ be any declared observable state such that every admissible primitive observation factors through α . Then every in-scope semantic property factors through α : there exists $\tilde{P}$ with*

$$P(c) = \tilde{P}(\alpha(c)) \quad \text{for all } c \in \mathcal{C}.$$

Proof. If every admissible primitive observation factors through α , then every admissible transcript factors through α , and so does any output computed from that transcript. ■

Corollary II.11 (Observable-state insufficiency). *If $\alpha(c_1) = \alpha(c_2)$ but $P(c_1) \neq P(c_2)$, then no admissible strategy can compute P with zero error without adding new information beyond the declared observable state.*

Remark II.12 (Interpretation). The sufficiency and insufficiency statements are the helper-view version of the information barrier: if a task varies inside one declared state fiber, that state is not sufficient and must be refined by additional information.

III. FIXED-LENGTH ZERO-ERROR SIDE INFORMATION

A. Fixed-Length Zero-Error Recovery

We now state the zero-error coding problem in generic representation-map form. Let C be a finite class variable and let $U = \pi(C)$ be the representation observed by the decoder. If the representation is non-injective on classes, then the decoder cannot determine C from U alone; the unresolved ambiguity is exactly the ambiguity inside the fibers of π .

Definition III.1 (Observation channel). Let C be a random class label, let $U = \pi(C)$ be its representation, and let the decoder observe U together with any charged auxiliary description supplied by the encoder.

Theorem III.2 (Zero auxiliary description iff injective representation). *Let $\mathcal{C}$ be a finite class space and let $\pi : \mathcal{C} \rightarrow \mathcal{U}$ be a representation map. Zero-error recovery of C from $U = \pi(C)$ with no auxiliary description is achievable if and only if π is injective on classes.*

(L: ACS8)

Proof. Achievability: If π is injective on classes, then observing $\pi(c)$ determines c uniquely. The decoder simply inverts the class-to-representation map.

Converse: Suppose two distinct classes $c_1 \neq c_2$ share the same representation value: $\pi(c_1) = \pi(c_2)$. Then any decoder $g(U)$ outputs the same class label on both inputs, so it cannot be correct for both. Hence zero-error recovery without auxiliary description is impossible. ■

In the finite explicit setting, the same condition can be read directly from the representation channel: one-symbol zero-error feasibility is equivalent to injectivity (GPH27). Interpreting π as a learned embedding, this is the exact threshold for zero-error class recovery from the representation alone.

Remark III.3 (Information-theoretic corollary). Under any distribution with positive mass on both colliding classes, $I(C; U) < H(C)$. This is an average-case consequence of the deterministic barrier above.

(L: ACS8)

Remark III.4 (Relation to Ahlswede-Dueck). In the identification paradigm of [6], the decoder asks “is the message m ?” rather than “what is the message?” This yields double-exponential codebook sizes. Our setting is different: we require zero-error identification of the *class*, not hypothesis testing. The one-shot zero-error identification feasibility threshold is binary rather than asymptotic.

(L: ACS8)

B. Fixed-Length Auxiliary Description

We now show that augmenting the representation with an auxiliary codeword restores exact recovery.

Definition III.5 (Auxiliary description). An *auxiliary description* is a value $M(c)$ associated with each class $c \in \mathcal{C}$ and transmitted in addition to the representation value $\pi(c)$. In the fixed-length setting, the auxiliary description alphabet has size 2^L .

(L: ACS7)

Theorem III.6 (Fixed-length sufficiency). *If $|\mathcal{C}| = k$, then an auxiliary description of length $L \geq \lceil \log_2 k \rceil$ bits suffices for zero-error identification, regardless of whether π is class-injective.*

(L: ACS7)

Proof. A class-injective auxiliary description assigns a unique identifier to each class. Reading that description together with the representation determines the class exactly. ■

C. Ambiguity Converse

The next theorem is the paper's main zero-error compression statement. It converts representation collision multiplicity directly into a necessary fixed-length side-information budget.

In graph terms, the present fixed-representation setting is a structured zero-error special case: confusable classes are exactly those lying in the same representation fiber, so the confusability graph is a disjoint union of cliques. The point of the section is therefore not a new general graph-entropy theorem, but the identification of the exact primitive governing this deterministic side-information regime and the theoremic consequences that follow from it.

Definition III.7 (Collision multiplicity). Let $\mathcal{C}$ be a finite class space and let $\pi : \mathcal{C} \rightarrow \mathcal{U}$ be a representation map. Define

$$A_\pi := \max_{u \in \text{Im}(\pi)} |\{c \in \mathcal{C} : \pi(c) = u\}|.$$

Thus A_π is the size of the largest class-collision block under the representation.

(L: LWDC1, LWDC3)

Remark III.8 (Computing the collision multiplicity). The converse is parameterized by A_π . When the class-profile map is given explicitly as a finite table, A_π is obtained by grouping classes by

profile and taking the largest fiber size. When π is given implicitly by a large program, circuit, or learned model, computing or approximating A_π becomes a separate representation-dependent problem. The theorem identifies the exact zero-error auxiliary description budget once A_π is known; it does not assume that estimating A_π is always tractable from an implicit description.

If one observation channel factors through a richer one, collision multiplicity cannot increase (GPH28). Thus adding observable structure can only lower the corresponding zero-error tag bound. Under the same representation reading, enriching the embedding cannot worsen zero-error separability.

Corollary III.9 (Coarsening a bottleneck cannot reduce residual coding cost). *Let π_{fine} and π_{coarse} be two representations on the same class domain, and assume*

$$\pi_{\text{coarse}} = f \circ \pi_{\text{fine}}$$

for some map f . Then

$$A_{\pi_{\text{fine}}} \leq A_{\pi_{\text{coarse}}}, \quad \log_2 A_{\pi_{\text{fine}}} \leq \log_2 A_{\pi_{\text{coarse}}}.$$

Moreover, for each fine representation value u ,

$$\lceil \log_2 |\pi_{\text{fine}}^{-1}(u)| \rceil \leq \lceil \log_2 |\pi_{\text{coarse}}^{-1}(f(u))| \rceil.$$

(L: GPH28, GPH37)

Proof. The first inequality is the monotonicity of collision multiplicity under factorization. Taking binary logarithms gives the worst-case fixed-length comparison. The pointwise adaptive comparison is the same monotonicity statement applied fiberwise before taking ceiling logarithms.

■

Definition III.10 (Maximal-Barrier Regime). The domain is *maximal-barrier* if $A_\pi = k$, i.e., all classes collide under the observation map.

(L: LWDC3)

Theorem III.11 (Converse). *For any classification domain, any fixed-length scheme achieving $D = 0$ requires*

$$2^L \geq A_\pi \quad \text{equivalently} \quad L \geq \log_2 A_\pi,$$

where A_π is the collision multiplicity.

(L: [LWDC1](#))

Proof. Fix a collision block $G \subseteq \mathcal{C}$ with $|G| = A_\pi$ and identical representation profile. For classes in G , the decoder's side information is identical, so zero-error decoding must separate those classes using auxiliary description outcomes. With L bits there are at most 2^L outcomes, hence $2^L \geq |G| = A_\pi$. ■

Remark III.12 (Coding interpretation). The converse is a lossless coding law for semantic identity. The relevant confusable alphabet is the largest representation-collision block, and zero-error separation of that block requires exactly enough auxiliary outcomes to name its members. The theorem therefore has the same fixed-length form as a classical zero-error side-information budget, but the governing alphabet size is induced by the representation rather than by the ambient class set alone.

Corollary III.13 (Maximal-Barrier Converse). *If the domain is maximal-barrier ($A_\pi = k$), any zero-error scheme satisfies $L \geq \log_2 k$.*

(L: [LWDC3](#))

In the running example, the collision block $\{A, B\}$ shows that $A_\pi = 2$, so Theorem [III.11](#) already forces $L \geq 1$ bit of auxiliary description for zero-error identification.

D. Exact Finite-Block Law

The ambiguity converse extends exactly to blocks, so the main compression law is not merely one-shot. The next theorem family gives exact finite-block scaling rather than an asymptotic approximation.

Theorem III.14 (Exact one-shot feasibility threshold). *For an alphabet of n auxiliary descriptions, zero-error class recovery is feasible if and only if*

$$A_\pi \leq n.$$

(L: [GPH13](#))

Proof. By definition, two classes are adjacent in the confusability graph exactly when they have the same representation profile and are distinct. Hence each nonempty profile fiber is a clique, and there are no edges between different fibers. The confusability graph is therefore a disjoint union of profile-fiber cliques. Zero-error auxiliary description is equivalent to a proper coloring of this graph, so an alphabet of size n is feasible exactly when each fiber can be colored injectively, i.e., exactly when every profile fiber has size at most n . Taking the maximum over fibers gives the criterion $A_\pi \leq n$. ■

Theorem III.15 (Exact block feasibility law). *For block length $t \geq 1$ with coordinatewise observation on $\mathcal{C}^t$, zero-error auxiliary description with alphabet size n_t is feasible if and only if*

$$A_\pi^t \leq n_t.$$

Equivalently, the minimal feasible block alphabet is exactly A_π^t .

(L: [GPH16-17](#), [GPH20](#))

Corollary III.16 (Linear log-bit scaling). *Let $L_t^* := \log_2(A_\pi^t)$ be the minimum block auxiliary-description budget in bits. Then*

$$L_t^* = t \log_2 A_\pi, \quad \frac{L_t^*}{t} = \log_2 A_\pi.$$

So per-entity tag rate is exactly stable across block length.

(L: [GPH18-19](#), [GPH24](#))

Remark III.17 (Why this block model). The block construction repeats the same semantic

identification task across t entities while keeping the representation map fixed and applying it coordinatewise. It is therefore the direct block analogue of repeated one-shot classification in this model: the collision structure multiplies across coordinates, and the required zero-error side-information budget scales accordingly.

In the running example, $A_\pi = 2$, so the exact block law gives $L_t^* = t$ bits.

IV. ADAPTIVE SIDE INFORMATION AND AVERAGE-CASE RATES

A. Representation-Adaptive Side Information

The fixed-length theorem treats all representation fibers through the worst case A_π . A finer coding problem allows the auxiliary description length to depend on the observed representation value u . Then the relevant budget is not the maximum fiber size but the size of the individual fiber $\pi^{-1}(u)$. This section therefore resolves the same residual ambiguity object at a finer scale: first fiberwise, then in source-weighted expectation, and finally through active binary clarification protocols.

Definition IV.1 (Representation-adaptive auxiliary description). An *adaptive auxiliary scheme* assigns to each representation value u a binary description budget $\ell(u)$ and a decoder that, given u together with a binary string of length at most $\ell(u)$, recovers the class exactly whenever $\pi(C) = u$.

Theorem IV.2 (Pointwise optimal adaptive bit budget). *For a representation fiber of size $|\pi^{-1}(u)|$, zero-error recovery under a binary auxiliary alphabet is feasible if and only if*

$$|\pi^{-1}(u)| \leq 2^{\ell(u)}.$$

Consequently, the pointwise optimal number of auxiliary bits on fiber u is

$$\ell^*(u) = \lceil \log_2 |\pi^{-1}(u)| \rceil.$$

(L: [GPH32](#))

Proof. Once u is fixed, the decoder only needs to distinguish classes inside the fiber $\pi^{-1}(u)$. With $\ell(u)$ binary bits there are at most $2^{\ell(u)}$ auxiliary outcomes, so exact recovery is possible exactly when that many outcomes suffice to name the fiber elements injectively. The minimal feasible budget is therefore the least integer whose power of two dominates the fiber size, namely $\lceil \log_2 |\pi^{-1}(u)| \rceil$. ■

Definition IV.3 (Expected adaptive description length). Let $U = \pi(C)$ be the induced representation random variable under a source law on classes. For an adaptive budget ℓ , define its expected auxiliary length by

$$\mathbb{E}[\ell(U)] = \sum_u P_U(u) \ell(u).$$

Theorem IV.4 (Adaptive expected-rate lower bound). *Any feasible adaptive auxiliary scheme satisfies*

$$\mathbb{E}[\ell(U)] \geq \mathbb{E}[\lceil \log_2 |\pi^{-1}(U)| \rceil].$$

Moreover, the pointwise optimal budget from Theorem IV.2 attains equality.

(L: [GPH32-33](#))

Proof. By Theorem IV.2, feasibility on fiber u requires

$$\ell(u) \geq \lceil \log_2 |\pi^{-1}(u)| \rceil$$

for every representation value u . Multiplying pointwise by $P_U(u)$ and summing over u gives the lower bound on expected length. Equality is achieved by choosing the pointwise optimal budget on every fiber. ■

Corollary IV.5 (Conditional-entropy lower bound). *Let C be drawn from any source distribution and let $U = \pi(C)$. Then any feasible adaptive auxiliary scheme satisfies*

$$\mathbb{E}[\ell(U)] \geq \frac{H(C | U)}{\log 2},$$

where $H(C | U)$ is conditional Shannon entropy in natural-log units.

(L: [GPH34](#))

Proof. For each representation value u , the conditional law of C given $U = u$ is supported on the fiber $\pi^{-1}(u)$, so its entropy is at most $\log |\pi^{-1}(u)|$. Averaging over u gives

$$H(C | U) \leq \mathbb{E}[\log |\pi^{-1}(U)|].$$

Since $\log |\pi^{-1}(u)| \leq \lceil \log_2 |\pi^{-1}(u)| \rceil \log 2$ fiberwise, Theorem [IV.4](#) yields the stated lower bound. ■

Theorem IV.6 (Fiberwise prefix-coding upper bound). *Assume standard prefix-coding achievability on each finite fiber. Then there exists an adaptive zero-error auxiliary code satisfying*

$$\mathbb{E}[\ell(U)] \leq \frac{H(C | U)}{\log 2} + 1.$$

(L: [GPH35](#))

Proof. Because the decoder already knows $U = u$, prefix-freeness is required only within each fiber $\pi^{-1}(u)$. Applying a standard prefix-coding theorem separately to the conditional source on each nonempty fiber gives an expected conditional code length at most $H(C | U = u)/\log 2 + 1$. Averaging over u yields the stated bound. ■

Corollary IV.7 (Zero-error conditional-entropy sandwich). *The optimal expected zero-error auxiliary rate lies between*

$$\frac{H(C | U)}{\log 2} \quad \text{and} \quad \frac{H(C | U)}{\log 2} + 1.$$

(L: [ZEC1](#))

Proof. The lower bound is Theorem [IV.4](#), and the upper bound is Theorem [IV.6](#). The two statements together give the claimed sandwich. ■

Corollary IV.8 (Active binary clarification). *If, after observing U , the decoder is allowed to ask adaptive binary clarification questions whose leaves identify the class exactly, then the minimum*

expected number of clarification questions is bounded between

$$\frac{H(C | U)}{\log 2} \quad \text{and} \quad \frac{H(C | U)}{\log 2} + 1.$$

(L: [GPH34-35](#))

Proof. Each binary clarification protocol induces a binary code over the conditional source on each fiber, and conversely any fiberwise binary prefix code can be implemented as an adaptive yes/no decision tree. The lower and upper bounds therefore follow from the preceding conditional-entropy comparison and fiberwise prefix-coding theorem. ■

Corollary IV.9 (Representation-respecting noisy transcripts cannot beat the fiber law). *Let T be any repeated or noisy observation transcript whose realized value factors through the fixed representation, i.e., $T = h(U)$ for some map h . Then recovering C from T cannot require less worst-case or pointwise adaptive side information than recovering C from U itself.*

(L: [GPH28](#), [GPH37](#))

Proof. If the entire transcript factors through U , then it is a coarsening of the original representation. Corollary [III.9](#) applies directly: coarsening cannot reduce the largest collision fiber and cannot reduce the pointwise adaptive fiber budget. Thus repeated or noisy observations that never break the original fibers cannot remove the residual zero-error coding burden. ■

Remark IV.10 (Coding interpretation). Theorem [IV.4](#) is the average-case counterpart of the max-fiber law. The fixed-length theorem pays for the worst collision fiber; the adaptive theorem pays for each fiber separately and averages under the source-induced law of U . Thus the fixed-length law, the pointwise fiber law, and the entropy sandwich are three resolutions of the same object: the cost of naming the classes that remain collapsed inside each representation fiber.

Remark IV.11 (Entropy comparison). The conditional-entropy lower and upper bounds place the adaptive theorem inside classical source-coding language: zero-error adaptive side information is sandwiched around $H(C | U)$ up to the standard one-bit prefix-coding overhead, but the exact

zero-error rate is still controlled fiberwise by finite collision structure rather than by entropy alone.

B. The Deterministic Corner: Semantic Identity Rate-Distortion on a Collision Block

The zero-error threshold does not exhaust the geometry. We now characterize the exact distortion-bits tradeoff in the cleanest deterministic setting: a uniform source on a single collision block. This isolates collision geometry alone, separate from source statistics and cross-fiber interactions. In that setting, the same collision structure determines a closed-form distortion curve as a function of the auxiliary budget.

Theorem IV.12 (Semantic identity rate-distortion theorem). *Let $G \subseteq \mathcal{C}$ be a collision block of size a , so all classes in G have the same representation profile. Assume the source class C is uniform on G . Then the optimal distortion under a fixed-length auxiliary budget of L bits is exactly*

$$D^*(L) = \max\left(0, 1 - \frac{2^L}{a}\right).$$

Equivalently, every fixed-length scheme with auxiliary budget L bits satisfies

$$\Pr[\hat{C} = C] \leq \frac{2^L}{a}, \quad D = \Pr[\hat{C} \neq C] \geq 1 - \frac{2^L}{a}.$$

If $2^L \geq a$, then zero distortion is achievable; if $2^L < a$, then the lower bound is tight.

(L: [GPH30](#))

Proof. Fix a realization of all internal randomness used by the scheme. Because every class in G has the same representation profile, the decoder's side information is the same on all classes in G ; on that block, the decoder can therefore distinguish classes only through the auxiliary outcome. With L bits there are at most 2^L outcomes, and for each such outcome the decoder can be correct on at most one class in G . Hence, for the fixed randomness, the scheme is correct on at most 2^L of the a classes in G , so its success probability under the uniform distribution on G is at most $2^L/a$. Averaging over the scheme's randomness preserves the same bound. The distortion lower bound is the complement of the success bound.

If $2^L \geq a$, assign distinct auxiliary outcomes to the classes in G and decode injectively; this gives zero distortion. If $2^L < a$, choose 2^L classes in G , assign them distinct auxiliary outcomes, and map all remaining classes to arbitrary auxiliary outcomes. Then the decoder is correct on exactly 2^L of the a classes, so the success probability is $2^L/a$ and the distortion is exactly $1 - 2^L/a$. This matches the converse. ■

Corollary IV.13 (Zero-error threshold). *Under the hypotheses of Theorem IV.12, zero distortion is achievable if and only if*

$$L \geq \log_2 a.$$

Proof. By Theorem IV.12, zero distortion is achievable exactly when $D^*(L) = 0$, equivalently when $2^L \geq a$, which is the same as $L \geq \log_2 a$. ■

Corollary IV.14 (Zero-budget error floor). *If $L = 0$ and the source is uniform on a collision block of size a , then any scheme relying only on the representation satisfies*

$$D \geq 1 - \frac{1}{a}.$$

In particular, if the source is uniform on a worst collision block with $a = A_\pi$, then

$$D \geq 1 - \frac{1}{A_\pi}.$$

(L: [GPH30](#))

Proof. Set $L = 0$ in Theorem IV.12. Since $2^0 = 1$, the distortion lower bound becomes $1 - 1/a$. Choosing a worst collision block gives the second statement. ■

Remark IV.15 (The cost of omission). This corollary quantifies the price of relying exclusively on semantic attributes. If one chooses not to store the auxiliary identifier required by the fixed-length coding law, the saved $\lceil \log_2 A_\pi \rceil$ bits are exchanged for a nonzero error floor. On a collision fiber of size 2, omitting the one-bit tag forces an error rate of at least 50%. On a collision fiber of size 100, omitting the seven-bit tag forces an error rate of at least 99%. One pays for identity

either in stored bits or in distortion. There is no third option in which identity is both free and exact.

Remark IV.16 (Why this matters). The distortion theorem is not only a converse to the zero-error law. It identifies the operational cost of omission in semantics-aware compression systems. If a system declines to store the identity bits needed to separate collided fibers, then the penalty is not abstract suboptimality but a concrete minimum error rate. The important feature is the steepness of the curve: because distortion scales as $1 - 2^L/a$, small deficits in the coding budget can produce large losses in identity accuracy. This is the exact storage-versus-distortion tradeoff induced by semantic abstraction.

Remark IV.17 (Steepness). The steepness is easy to see numerically. On a collision block of size 100, zero distortion requires at least $\lceil \log_2 100 \rceil = 7$ bits. With only 6 bits the distortion is at least 36%, and with only 5 bits it is at least 68%. Small shortfalls below the exact-recovery threshold can therefore make identity accuracy unusable in practice.

Remark IV.18 (Scope of the exact curve). The exact formula above is stated for the uniform source on a single collision block. That restriction is intentional. It isolates a deterministic corner in which collision geometry alone determines the full tradeoff. For non-uniform sources on the same block, or for mixtures across multiple fibers, the optimal distortion curve depends on the source law as well as the fiber sizes; in those settings one expects a different shape, closer in spirit to entropy-weighted side-information bounds than to the uniform worst-block law treated here. The point of the present theorem is exact isolation of the geometry-only effect.

Proposition IV.19 (Finite fiberwise decomposition). *Fix a uniform per-fiber auxiliary alphabet of size T . Among all finite subsets S of classes satisfying $|S \cap \pi^{-1}(u)| \leq T$ for every representation value u , the maximum recoverable source mass equals the sum of the fiberwise maxima. Moreover, this optimum is attained by selecting, independently in each fiber, a subset of size at most T with maximum source mass and then taking the union across fibers.*

(L: [FRD1-2](#))

Proof. Any feasible subset decomposes into its slices inside the representation fibers, and its total source mass is the sum of the masses of those slices. Each slice contributes at most the maximum source mass achievable in its own fiber under the cardinality bound T , so the global recoverable mass is bounded above by the sum of the fiberwise optima. Conversely, choosing a maximizing subset independently inside each fiber and taking their union preserves the per-fiber cardinality bound and attains equality. ■

C. Worked Example

Return to the running example with profiles

$$A = 000, \quad B = 000, \quad C = 100, \quad D = 110, \quad E = 111.$$

The fixed-length theorem gives $A_\pi = 2$, so one extra bit is necessary and sufficient for exact recovery. The adaptive theorem says more: the fiber over profile 000 requires one bit, while the singleton fibers over 100, 110, and 111 require zero bits. Thus the expected auxiliary rate is exactly the probability of landing in the collided fiber.

This is the representation-level interpretation the paper emphasizes. The representation already resolves three of the four fibers completely; the only irreducible metadata cost is the one residual collision between A and B .

The next section turns this coding object into a representation-sufficiency theory: once the residual ambiguity of each fiber is explicit, one can ask exactly which downstream tasks, helper views, and modular factors eliminate it and which do not.

D. Secondary Query and Frontier View

The paper also studies a query resource W for settings where the decoder is allowed to interrogate primitive attributes rather than receive all auxiliary information up front. That query-side viewpoint is secondary here: the main compression object is the side-information rate needed to resolve representation collisions. We retain the query-side results later because they show one alternative way to pay for the same unresolved ambiguity.

V. REPRESENTATION SUFFICIENCY, AUXILIARY VIEWS, AND MODULAR STRUCTURE

a) Why semantic representation alone cannot distinguish.: The mechanization proves that semantic representation cannot distinguish entities with identical observable profiles (ACS8), cannot track provenance or source identity (ACS9), and cannot embed explicit identity information into the semantic layer (EMB1). These impossibility theorems establish that the information barrier is not a limitation of specific encodings but a fundamental property of representation-level observation. In compression-theoretic terms, semantic representation alone is a *lossy* identity compressor. When distinct entities map to the same semantic representation, no decoder can recover the original identity without additional side information (NSL2, NSL3).

The coding laws above quantify how much ambiguity a fixed representation leaves unresolved. We now turn that same fiber geometry into a representation-sufficiency theory: which downstream tasks factor through the representation, when auxiliary views genuinely help, and when modular latent structure adds exact information rather than architectural redundancy.

The fixed-length law measures the worst unresolved collision,

$$L_{\text{fix}}^* = \log_2 A_\pi,$$

while the adaptive law measures the source-weighted cost of the individual fibers,

$$L_{\text{adapt}}^* = \mathbb{E}[\lceil \log_2 |\pi^{-1}(U)| \rceil].$$

Thus representation collisions translate directly into auxiliary metadata cost. Injective representations need no additional description; noninjective ones require either extra side information or nonzero error.

A. Instantiations: Learned Representations and Formal Witnesses

The abstract coding framework applies to any representation map. We begin with the learned-representation setting that motivates many modern systems, and then turn to a formal entity-attribute model that gives a machine-checked formal instance of the same laws.

The same fiber picture clarifies the distinction between semantic description and exact identity. Semantic information is what the representation exposes. Identity information is the exact class label or entity name that may still need to be carried explicitly. Zero-error recovery from semantic information alone is possible exactly when the semantic representation is injective on identity classes. When it is not, the collision fibers quantify the irreducible identity residue that must be restored by explicit metadata.

This transfers directly to neuro-symbolic systems. A neural encoder supplies the semantic side by producing a learned representation; a symbolic layer attempts to recover or reason about exact class-level distinctions. The fiber geometry says precisely when this is possible without extra scaffolding and when it is not: symbolic reasoning cannot recover distinctions already collapsed by the encoder, except by receiving additional identifiers, additional observables, or additional disclosures.

a) Why symbols are necessary for systems with semantics.: Semantic representations derive their utility from abstraction: they compress varied instances into shared categories, concepts, or embedding regions. That abstraction is typically non-injective. It creates semantic fibers in which distinct identities are deliberately conflated. The coding laws then force an exact consequence: recovering identity from a fiber $\pi^{-1}(u)$ requires an auxiliary description of $\lceil \log_2 |\pi^{-1}(u)| \rceil$ bits, and in the worst case requires $\log_2 A_\pi$ bits by Theorem III.11. A system therefore needs a mechanism to supply those bits. In computational systems, that mechanism is usually a symbolic handle: a pointer, a unique identifier, a variable name, a primary key, or an index entry.

This clarifies the theoremic and systems-level claims. The theorem says that injective auxiliary information is necessary. The systems implication is that symbolic handles are the standard way real systems implement that injective auxiliary channel. In that sense, the neuro component provides the semantic compression, while the symbolic component pays the identity-bit budget mandated by the coding laws.

b) What this means for ML systems.: The instantiated theorems have immediate implications for learned representation systems.

- **Concept bottleneck models.** If the intermediate concept representation has collisions,

then no downstream classifier can distinguish the collapsed concepts without extra identity information. Operationally, exact concept-level decisions require an added disambiguation channel, a retrieval key, or an explicit symbolic concept identifier.

- **Retrieval-augmented systems.** The retrieval index pays the identity bit budget. If two artifacts have the same embedding, the system must store distinguishing metadata elsewhere, such as document identifiers, source pointers, or exact lookup keys. Otherwise exact retrieval is impossible on the collided fiber.
- **Pure representation schemes.** Systems that attempt to avoid any nominal layer do not escape the theorem. They can still be useful, but they cannot guarantee zero-error open-world identity when the representation is non-injective.
- **Open-world deployments.** Any system that must identify entities exactly in an open world must either use an injective encoder or carry explicit identity metadata. There is no third option.

c) Systems-level reading.: The coding laws clarify several common design patterns. Pure embedding retrieval without identifiers cannot guarantee exact retrieval on collided fibers; if two documents share the same embedding, distinguishing metadata must be stored somewhere else. Concept bottleneck systems without concept-level identifiers cannot recover which ground-truth instance generated a collapsed concept representation. Semantic caches keyed only by semantic hashes can preserve approximate meaning while still returning the wrong identity when collisions occur. In each case, the coding laws identify what additional information restores exactness, and what distortion floor is inherited if that information is omitted.

Semantic representation alone is lossy identity compression. The auxiliary identity channel restores the missing distinguishing information. In deployed systems, that information may appear as identifiers, registry entries, retrieval keys, or explicit symbolic names, but it must appear somewhere if zero-error identity is the goal.

The same necessity extends beyond neural models. Compilers use symbol tables because semantic structure alone does not distinguish different bindings of the same name. Databases use primary keys because attribute descriptions can collide across records. Retrieval systems use

document identifiers because embeddings can collide across sources. The contribution of this paper is to quantify that systems-level identity deficit exactly: $\log_2 A_\pi$ bits in the worst case, or $\lceil \log_2 |\pi^{-1}(u)| \rceil$ bits on fiber u .

d) Attribute-only versus tag-augmented design.: The coding laws imply a sharp system-design dichotomy whenever $A_\pi > 1$. In an attribute-only strategy, the system stores or transmits only the semantic representation $\pi(v)$. This pays only for the semantic description, but it inherits the distortion floor forced by the collision fibers; in the zero-budget case, the worst collided block incurs error at least $1 - 1/A_\pi$ by Corollary IV.14. In a tag-augmented strategy, the system stores $\pi(v)$ together with auxiliary identity information. This increases storage by the exact identity budget, up to $\lceil \log_2 A_\pi \rceil$ bits in the worst case, but restores zero-error identity recovery. Operationally, the semantic layer handles the “what,” while the auxiliary tag handles the “which.”

This tradeoff explains the ubiquity of symbolic identifiers in high-reliability systems. Primary keys, memory addresses, retrieval identifiers, and similar handles are not mere conveniences; they are the mechanisms by which systems avoid the distortion floor mandated by the coding laws. If the application requires exact retrieval, verification, provenance tracking, or entity-level consistency, the attribute-only strategy is infeasible and the tag-augmented strategy is mandatory.

e) Formal instantiation.: The paper also develops a concrete formal identity domain in Lean. In that model, entities carry an explicit attribute family $\mathcal{I}$, primitive observables are the associated binary maps q_I , and the representation is the induced profile map $\pi(v) = (q_I(v))_{I \in \mathcal{I}}$. In this setting, representation-computable properties reduce to attribute-computable ones, and the generic coding laws specialize to a machine-checked formal instance.

The formal theorems establish that equal observable profiles cannot distinguish identity (ACS8), that provenance is not recoverable from the profile alone when collisions exist (ACS9), and that nominal identity cannot be embedded back into the semantic layer (EMB1). Readers interested only in the information-theoretic content may treat this development as a rigorous formal instantiation rather than as a prerequisite for the main theorem arc.

f) Instantiated neurosymbolic linking theorems.: Within that formal identity domain, the generic helper-view theorems specialize by setting the task to identity $Y = C$. The mechanization

(NSL1–NSL6) proves the following instantiated claims.

- If the symbolic handle σ is injective, the neurosymbolic link (π, σ) achieves zero-error identity recovery (NSL1). This is the identity-task instantiation of the injectivity criterion GPH25.
- If π has collisions, semantic representation alone cannot achieve zero-error identity (NSL2), and no decoder can recover the collapsed identities without identity-resolving side information (NSL3).
- The exact characterization is: (π, σ) recovers identity if and only if σ distinguishes all entities within each semantic fiber (NSL4). An injective handle therefore suffices (NSL5). This is the identity-task instantiation of the helper-view theorem GPH40.
- The pair (π, σ) is the canonical helper view for open-world identity systems (NSL6).

These instantiated theorems connect the generic information-theoretic helper-view framework to a concrete formal identity domain. Together with ACS8, ACS9, and EMB1, they show how the auxiliary identity channel pays for the ambiguity that the semantic layer leaves unresolved.

B. Task Sufficiency, Helper Views, and Factorization

The neurosymbolic reading is one specialization of a broader theorem family. The same fiber geometry characterizes when a representation is sufficient for an arbitrary downstream task, when helper views genuinely help, and when factorized modules contribute real exact information.

Definition V.1 (Task-fiber ambiguity). Let $Y = f(C)$ be a downstream task label induced by the class. For a fixed representation π , define

$$A_{\pi \rightarrow Y} := \max_u |\{f(c) : \pi(c) = u\}|.$$

This is the maximum number of task labels that remain unresolved inside one representation fiber.

Theorem V.2 (Task sufficiency criterion). *The representation alone supports zero-error recovery of Y if and only if $A_{\pi \rightarrow Y} \leq 1$.*

(L: GPH38)

Proof. Zero-error recovery of Y from U alone is possible exactly when every representation fiber carries a single task label. That condition is equivalent to the statement that the largest task-fiber ambiguity is at most one. ■

Corollary V.3 (Task-level coarsening monotonicity). *If $\pi_{\text{coarse}} = g \circ \pi_{\text{fine}}$, then*

$$A_{\pi_{\text{fine}} \rightarrow Y} \leq A_{\pi_{\text{coarse}} \rightarrow Y}.$$

In particular, coarsening a representation cannot improve exact downstream task sufficiency.

(L: GPH39)

Proof. Every coarse fiber is a union of fine fibers, so collapsing from π_{fine} to π_{coarse} cannot reduce the number of task labels realizable inside the worst fiber. ■

Theorem V.4 (Helper-view sufficiency). *Let $S = \sigma(C)$ be an additional deterministic view available to the decoder. Then exact recovery of Y from (U, S) is possible if and only if each joint fiber of (π, σ) carries at most one task label.*

(L: GPH40)

Proof. The pair (U, S) is just a refined deterministic representation. Exact recovery is therefore possible exactly when the downstream task is constant on the fibers of that refined representation. ■

Corollary V.5 (Helper-view monotonicity). *Adding a deterministic helper view cannot worsen task ambiguity. In particular, the task-fiber ambiguity under (U, S) is at most the task-fiber ambiguity under U alone.*

(L: GPH41)

Proof. Passing from U to (U, S) refines the representation fibers, so the largest number of task labels per fiber can only decrease. ■

Corollary V.6 (Factorized product law). *In a genuinely factorized product setting, where the class space decomposes as a product and the representation decomposes coordinatewise, the joint fiber is the product of the component fibers and the worst-case residual ambiguity multiplies exactly across modules.*

(L: [GPH42-43](#))

Proof. For coordinatewise product representations, each joint fiber is exactly the product of the corresponding component fibers, so their cardinalities multiply. Taking the largest such product yields multiplicativity of the worst-case residual ambiguity. ■

For learned representations, estimating quantities such as A_π , $A_{\pi \rightarrow Y}$, or the adaptive fiber profile is itself a representation-analysis problem rather than part of the theorem. In a finite explicit registry or index this is just fiber counting; for a large learned embedding one would need either exact collision enumeration after discretization or provable upper and lower bounds on the induced fibers. The point of the present results is to identify what those quantities mean once they are known.

Read concretely, the task-sufficiency theorem is the exact-recovery analogue of a sufficient-statistic condition for the downstream task Y : the representation is sufficient precisely when every fiber carries one task label. The helper-view and factorized laws then say when concept bottlenecks, retrieval-augmented pipelines, multimodal encoders, or modular neural architectures contribute genuinely new exact information rather than repackaging ambiguity already present in the base representation.

In the finite explicit setting, one-symbol zero-error feasibility is equivalent to injectivity and refinement cannot increase collision multiplicity ([GPH27](#), [GPH28](#)). The same setting also yields a disclosure separation between nominal access and tag-free identity resolution ([PRIV3](#)).

C. One Additional Semantic-Identification Example

Consider a model registry in which artifacts are indexed by a semantic representation π built from declared capabilities or structural signatures. If two distinct artifacts have the same

representation value, then the representation alone cannot support exact registry identification. The fixed-length theorem says that the worst-case auxiliary identifier length is determined by the largest collision bucket. The adaptive theorem says that rare collision buckets contribute little to expected metadata cost, while large or frequent buckets dominate it.

This example is close to the intended task-aware compression reading: the representation is already useful for search and organization, but exact identification may still require additional stored metadata. The theorem identifies the exact price of that residual ambiguity.

VI. QUERY COSTS OF RESIDUAL AMBIGUITY

A. Query Families and Distinguishing Sets

We now record the query-side version of the same representation-sufficiency problem. The preceding sections quantify how many auxiliary bits are needed to resolve representation collisions directly. An alternative operational response to the same residual ambiguity is to spend query resources rather than transmission resources: instead of sending side information to separate a fiber, one can interrogate additional primitive attributes until the class is determined. Equivalently, a query family induces an additional deterministic helper view, and the question is how large that helper view must be before exact recovery becomes possible. A distinguishing set separates every pair of distinct classes. In full generality, inclusion-minimal distinguishing sets need not have the same size, so the first invariant we use is the minimum number of primitive queries needed for zero-error separation.

This section is secondary to the main side-information results, but it belongs to the same second arc. Its role is to quantify how much adaptive interrogation is needed when one chooses not to send the required side information up front. In that sense, the minimum distinguishing number d is the minimum size of a sufficiency-restoring helper view built from primitive observables. It is to queries what fiber size is to side-information bits: both measure how much unresolved ambiguity remains after the representation is fixed.

Definition VI.1 (Query family). Let $\mathcal{Q}$ be the set of all primitive queries available to an observer. For a classification system with attribute set $\mathcal{I}$, we have $\mathcal{Q} = \{q_I : I \in \mathcal{I}\}$ where $q_I(v) = 1$ iff

v satisfies attribute I .

(L: [ACSI](#))

In this section, “queries” means primitive attribute predicates from $\Phi = \mathcal{Q}$.

Definition VI.2 (Distinguishing set). A subset $S \subseteq \mathcal{Q}$ is *distinguishing* if, for all values v, w with $\text{class}(v) \neq \text{class}(w)$, there exists $q \in S$ such that $q(v) \neq q(w)$.

Definition VI.3 (Minimum distinguishing number). The *minimum distinguishing number* of a classification system is

$$d := \min\{|S| : S \subseteq \mathcal{Q} \text{ is distinguishing}\}.$$

Equivalently, d is the smallest number of primitive queries that suffices to separate all classes with zero error.

Remark VI.4 (Inclusion-minimal versus minimum-cardinality). These notions differ: an inclusion-minimal distinguishing set may still be larger than a minimum-cardinality one. The lower bound below depends only on the operational quantity d .

In the running example from Section [III](#), $\{q_1, q_2, q_3\}$ is distinguishing and no pair of primitive queries is distinguishing, so the minimum distinguishing number is $d = 3$.

B. Implications for Witness Cost

Theorem VI.5 (Attribute-only lower bound). *For attribute-only observers, class identity checking requires*

$$W(\text{class-identity}) = \Omega(d)$$

in the worst case, where d is the minimum distinguishing number.

(L: [ACSI](#))

Proof. Assume a zero-error attribute-only procedure halts after fewer than d queries on every execution path. Fix any execution path and let $Q \subseteq \mathcal{I}$ be the set of queried attributes on that

path, so $|Q| < d$. By definition of d , no set of size $< d$ is distinguishing; hence there exist values v, w from different classes with identical answers on all attributes in Q .

An adversary can answer the procedure's queries consistently with both v and w along this path. Therefore the resulting transcript (and output) is identical on v and w , contradicting zero-error class identification. So some execution path must use at least d queries, giving worst-case cost $\Omega(d)$. ■

Corollary VI.6 (Lower bound on attribute-only witness cost). *For any attribute-only observer, $W(\text{class-identity}) \geq d$ where d is the minimum distinguishing number.*

(L: [ACSI](#))

In the running example, this yields $W \geq 3$ for any attribute-only zero-error scheme, even though the ambiguity converse will later show that one explicit tag bit already suffices to resolve the only collision block.

This is the complementary resource law. The ambiguity converse says how many bits are needed to eliminate collision multiplicity; the query lower bound says how much additional structure must still be traversed when those bits are absent. Read through the second arc, the bit law measures the residual description cost of restoring sufficiency, while the query law measures the minimal observable augmentation needed to restore it operationally. When side information is cheap but interrogation is costly, the fiber-size bound governs; when transmission is expensive but queries are available, the distinguishing-number bound governs.

C. Structured Query Families and Matroid Behavior

The general attribute-query model above does not force inclusion-minimal distinguishing sets to be equicardinal. A finite explicit counterexample is available even for four classes and four primitive binary queries ([GPH31](#)). A stronger structural theorem becomes valid only after one restricts the query family. In that restricted setting, the question becomes: when primitive observables are treated as irreducible helper-view coordinates, what is the structure of the minimal coordinate families that restore exact recoverability?

Definition VI.7 (Structured axis model). A *structured axis model* is a query family presented together with an explicit derivability relation between primitive observables, so that one query may semantically subsume another. A query family is *orthogonal* if no primitive observable is derivable from a distinct primitive observable in the family.

Remark VI.8 (Why this restriction is coherent). If the observable family contains redundant or semantically derivable coordinates, then inclusion-minimality depends on representational accidents. The structured axis model removes that artifact by quotienting out derivable observables and restricting attention to orthogonal primitive axes.

Theorem VI.9 (Restricted matroid theorem for structured axes). *In the structured axis model, semantically minimal complete axis families form the bases of a matroid. Consequently, all such families have equal cardinality.*

(L: [L1](#), [L5](#))

Proof. The formal theorem is proved for the axis-based representation in which semantic minimality implies orthogonality, orthogonality implies exchange, and standard matroid theory then gives equicardinality of maximal independent sets. The key point is that the theorem is about structured primitive axes modulo derivability, not about arbitrary subsets of binary class-indicator queries. ■

Remark VI.10 (Relation to the general query invariant). The restricted matroid theorem strengthens the general lower-bound story when the primitive queries come from a derivability-aware axis presentation. In the unrestricted attribute-query model, the invariant used in the paper is the minimum distinguishing number d ; in the structured axis model, the same quantity is certified by matroid basis cardinality.

Observer Class	Witness Procedure	Witness Cost W
Nominal-tag	Single tag read	$O(1)$
Attribute-only	Query a distinguishing family of minimum size	$\Omega(d)$

TABLE I
WITNESS COST FOR CLASS IDENTITY BY OBSERVER CLASS.

D. Witness Cost Comparison

VII. ROBUSTNESS UNDER SYSTEM GROWTH

A. Secondary Robustness and Decidability Boundary

The main side-information theorems address whatever collision structure is present in the realized representation. A complementary question is whether that fiber geometry is stable under system evolution: can a representation that is currently collision-free become noninjective when the class universe grows? The next two results are secondary robustness statements quantifying that fragility.

Definition VII.1 (Finite realized world). Let $\mathcal{C}$ be the class universe. A *finite realized world* is a finite subset $W \subseteq \mathcal{C}$ of classes currently present in a deployed system snapshot. We call W *barrier-free* iff

$$\forall c_1, c_2 \in W, \pi_{\mathcal{C}}(c_1) = \pi_{\mathcal{C}}(c_2) \Rightarrow c_1 = c_2.$$

(L: [ROB1](#))

Definition VII.2 (Open-world extension). An *open-world extension* is a super-world $W' \supseteq W$ obtained by adding classes while keeping the observation family Φ fixed.

(L: [ROB1-2](#))

Theorem VII.3 (Barrier-freedom is not extension-stable). *In the open-world class model used here, it is not true that barrier-freedom is preserved under all extensions:*

$$\neg \left(\forall W \subseteq W', \text{BarrierFree}(W) \Rightarrow \text{BarrierFree}(W') \right).$$

(L: *ROBI-2*)

Proof. Take the empty world $W_0 = \emptyset$, which is barrier-free. In this model, two concrete classes are shape-equivalent (same observable profile) but nominally distinct, so adding them yields an extension $W_1 \supseteq W_0$ that is not barrier-free. Therefore barrier-freeness is not extension-stable.

■

Remark VII.4 (Safety interpretation). The theorem formalizes a basic fact about open-world systems: present collision-freeness does not by itself yield a stable future guarantee. Persistent zero-error claims therefore need an explicit growth model rather than a one-time snapshot argument.

The running example offers the same intuition in finite form: the realized world $\{A, C, D, E\}$ is collision-free, but adding class B immediately restores the collision block $\{A, B\}$.

For compression system design, the point is direct: a representation that currently needs no auxiliary identification bits can require a positive residual coding budget after extension if new classes enter existing fibers. Thus open-world growth can change the optimal exact side-information rate even when nothing about the deployed decoder changes.

Definition VII.5 (Function-level barrier predicate). For a partial observer generator $f : \mathbb{N} \rightharpoonup \mathbb{N}$, define

$$\text{HasBarrier}(f) :\Leftrightarrow \exists x \neq y, \exists z, z \in f(x) \wedge z \in f(y).$$

Define $\text{BarrierFree}(f) :\Leftrightarrow \neg \text{HasBarrier}(f)$.

(L: *UNDI-2*)

Theorem VII.6 (Rice-style non-computability of barrier certification). *There is no computable predicate on program codes deciding whether the generated observer has a barrier:*

$$\neg \text{ComputablePred}(c \mapsto \text{HasBarrier}(\text{eval}(c))).$$

Equivalently, barrier-freeness certification is also non-computable.

(L: [UND1-2](#))

Proof. Assume there is a computable certifier for $P(c) := \text{HasBarrier}(\text{eval}(c))$. The property is non-trivial: some generators produce collisions and some do not. Rice’s theorem therefore implies that if membership in the corresponding semantic class were computable from code, every partial recursive function would have to satisfy it, contradiction. The barrier-free statement is equivalent by negation. ■

Remark VII.7 (Certification interpretation). The Rice-style theorem is the algorithmic counterpart to extension instability. Even when a current deployment appears collision-free, there is no general computable procedure that certifies this property for arbitrary future-generating code. Formal guarantees therefore require either a restricted extension model or explicit identity-carrying information that remains valid under growth.

Corollary VII.8 (Certification consequence for persistent $D = 0$ claims). *For unrestricted open-world generators, one cannot computably certify that attribute-only identification remains barrier-free under all future extensions. Consequently, persistent $D = 0$ guarantees cannot rely on “currently collision-free” attribute structure alone.*

(L: [ROB1-2](#), [UND1-2](#))

VIII. CONCLUSION

This paper studies what remains to be coded once a semantic representation is fixed. The answer is residual ambiguity: the collision fibers of the representation determine the exact cost of recovering identity. In the zero-error regime, that cost is the semantic identity rate $\log_2 A_\pi$. In the adaptive regime, it becomes the fiberwise budget $\lceil \log_2 |\pi^{-1}(u)| \rceil$. In the clean deterministic lossy corner, it becomes an exact distortion curve on a collision block.

The main conceptual point is that these are not different problems but different faces of the same one. If the representation is injective, identity is free. If it is non-injective, the missing information must be paid for somewhere. One may pay for it in auxiliary bits, in helper views, in additional disclosures, in query effort, or in distortion. The paper’s contribution is to make

that accounting exact. The later mechanized extensions sharpen this picture in three directions: finite multi-fiber recoverable mass decomposes fiberwise and is attained by independent per-fiber optimization (FRD1, FRD2); the zero-error expected-length problem is sandwiched around conditional entropy up to one bit (ZEC1); and simple growth or budget-allocation models can already be phrased as exact finite optimization problems (GRC1, GTB1, BST1).

The paper does not attempt a full stochastic rate-distortion theory. Its contribution is to solve a deterministic corner exactly and to show why that corner already governs many semantics-versus-identity design choices.

The broader payoff is representation sufficiency. The same ambiguity-fiber geometry governs when a representation is sufficient for exact downstream tasks, when helper views or retrieval-style augmentation genuinely help, and when modular or factorized latents add exact information rather than architectural redundancy. In the learned-representation reading, the interpretation is direct: injective representations support exact downstream identification with no extra metadata, while noninjective representations impose an irreducible side-information cost and, if that cost is left unpaid, a correspondingly irreducible distortion floor.

Query costs and robustness do not begin new stories; they record other ways the same unresolved ambiguity reappears when it is not paid for directly in side-information bits. One can transmit side information to resolve it, or instead build a sufficiently rich helper view out of primitive observables and pay for that view through interactive/query resources; in the structured-axis setting, the matroid law describes the geometry of such minimal sufficiency-restoring augmentations. One can also ask whether an apparently collision-free representation remains safe under world growth. These modalities are different ways of accounting for the same ambiguity left by a noninjective representation.

a) The neurosymbolic necessity theorem.: The central systems-level contribution is a formal justification for symbolic mechanisms in semantic systems. The coding laws prove that semantic abstraction imposes an exact identity cost: a non-injective representation requires auxiliary description to resolve its collision fibers, with worst-case cost $\log_2 A_\pi$ bits and fiberwise cost $\lceil \log_2 |\pi^{-1}(u)| \rceil$. Systems that require both semantic generalization and exact identity must

therefore implement a mechanism to pay this cost.

Symbolic handles, identifiers, pointers, keys, and index entries are standard computational instantiations of that payment. They are not the only abstract possibility, but they are common systems mechanisms. Consequently, the integration of symbolic identity layers into neurosymbolic, retrieval, and other semantics-aware systems follows naturally from the coding laws. The steepness of the distortion curve explains the practical importance: because distortion scales as $1 - 2^L/a$, even a small shortfall below the identity threshold can make exact identity unusable in practice. In the instantiated formal domain, the pair consisting of a neural encoder and an injective symbolic handle achieves zero-error identity recovery if and only if the handle distinguishes entities within each semantic fiber (NSL1–NSL6).

Acknowledgment: AI-use Disclosure

Generative AI tools (including Codex, Claude Code, Augment, Kilo, and OpenCode) were used throughout this manuscript, across all sections (Abstract, Introduction, theoretical development, proof sketches, applications, conclusion, and appendix) and across all stages from initial drafting to final revision. The tools were used for boilerplate generation, prose and notation refinement, LaTeX and structure cleanup, translation of informal proof ideas into candidate formal artifacts (Lean and LaTeX), and repeated adversarial reviewer-style critique passes to identify blind spots and clarity gaps.

The author retained full intellectual and editorial control, including problem selection, theorem statements, assumptions, novelty framing, acceptance criteria, and final inclusion or exclusion decisions. No technical claim was accepted solely from AI output. Formal claims reported as machine-verified were admitted only after Lean verification (no `sorry` in cited modules) and direct author review; Lean was used as an integrity gate for responsible AI-assisted research. The author is solely responsible for all statements, citations, and conclusions.

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Lean Handle Index.

ID	Handle	ID	Handle
ACS1	adversary_forces_n_minus_1_queries AbstractClassSystem/Extended.lean	ACS7	nominal_localization_constant_semantic AbstractClassSystem/Extended.lean
ACS8	shape_cannot_distinguish AbstractClassSystem/Core.lean	ACS9	shape_provenance_impossible AbstractClassSystem/Core.lean
BST1	Ssot.Paper1IT.exists_minimizing_split Paper1IT/BudgetSplit.lean	EMB1	semantic_non_embeddability AbstractClassSystem/Typing.lean
FRD1	ObserverModel.feasible_subsetMass_le_optimalFeasibleMass Paper1IT/FiberRateDistortion.lean	FRD2	ObserverModel.optimalSubset_attains_optimalFeasibleMass Paper1IT/FiberRateDistortion.lean
GPH13	Ssot.GraphEntropy.tagFeasible_iff_maxFiberCard_le	GPH16	Ssot.GraphEntropy.block_tagFeasible_iff_pow_maxFiberCard_le Paper1IT/GraphEntropyAsymptotic.lean
GPH17	Ssot.GraphEntropy.maxFiberCard_blockObserve_eq_pow Paper1IT/GraphEntropyAsymptotic.lean	GPH18	Ssot.GraphEntropy.blockTagRateBits_eq_mul_oneShot Paper1IT/GraphEntropyAsymptotic.lean
GPH19	Ssot.GraphEntropy.blockTagRateBitsPerCoordinate_eq_oneShot Paper1IT/GraphEntropyAsymptotic.lean	GPH20	Ssot.GraphEntropy.minBlockFeasibleAlphabet_eq_pow Paper1IT/GraphEntropyAsymptotic.lean
GPH24	Ssot.GraphEntropy.feasibleAtAlphabetBase_iff Paper1IT/GraphEntropyAsymptotic.lean	GPH25	Ssot.GraphEntropy.maxFiberCard_le_one_of_injective
GPH27	Ssot.GraphEntropy.tagFeasible_one_iff_injective	GPH28	Ssot.GraphEntropy.maxFiberCard_mono_of_factors_through
GPH30	Ssot.GraphEntropy.uniformFiberErrorRate_ge_one_sub_tag_ratio	GPH31	Ssot.GraphEntropy.inclusionMinimalCoordinateDistinguishing_not_equicardinal
GPH32	Ssot.GraphEntropy.optimalFiberBitLength_feasible	GPH33	Ssot.GraphEntropy.optimalExpectedAdaptiveBitLength_le
GPH34	Ssot.GraphEntropy.conditionalEntropyGiven_le_log2_mul_expectedAdaptiveBitLength	GPH35	Ssot.GraphEntropy.exists_conditionalCodes_expectedLength_le_entropy_bits_plus_one
GPH37	Ssot.GraphEntropy.optimalFiberBitLength_mono_of_factors_through	GPH38	Ssot.GraphEntropy.taskRecoverable_iff_maxTaskFiberCard_le_one
GPH39	Ssot.GraphEntropy.maxTaskFiberCard_mono_of_factors_through	GPH40	Ssot.GraphEntropy.taskRecoverable_pair_iff
GPH41	Ssot.GraphEntropy.maxTaskFiberCard_pair_le_left	GPH42	Ssot.GraphEntropy.observeFiber_prod_card
GPH43	Ssot.GraphEntropy.maxFiberCard_prod	GRC1	Ssot.Paper1IT.poissonCellCollisionProb_eq_one_sub_exp_mul_one_add Paper1IT/GrowthCollisions.lean
GTB1	Ssot.Paper1IT.requiredTagBits_positive_iff_two_le Paper1IT/GrowthTagBudget.lean	L1	matroid_basis_equicardinality axis_framework.lean
L5	I5_exchange_wrapper HandleAliases.lean	LWDC1	LWDConverse.collision_block_requires_bits lwd_converse.lean
LWDC3	LWDConverse.maximal_barrier_requires_bits lwd_converse.lean	NSL1	AbstractClassSystem.Neurosymbolic.neurosymbolic_zero_error_identity AbstractClassSystem/Neurosymbolic.lean
NSL2	AbstractClassSystem.Neurosymbolic.shape_alone_not_zero_error AbstractClassSystem/Neurosymbolic.lean	NSL3	AbstractClassSystem.Neurosymbolic.neurosymbolic_necessary_for_zero_error AbstractClassSystem/Neurosymbolic.lean
NSL4	AbstractClassSystem.Neurosymbolic.neurosymbolic_identity_recovery AbstractClassSystem/Neurosymbolic.lean	NSL5	AbstractClassSystem.Neurosymbolic.neurosymbolic_injective_implies_recoverable AbstractClassSystem/Neurosymbolic.lean
NSL6	AbstractClassSystem.Neurosymbolic.neurosymbolic_is_canonical_helper_view AbstractClassSystem/Neurosymbolic.lean	OM1	ObserverModel.finiteRateDistortionConverse Paper1IT/FiniteRateDistortionConverse.lean

ID	Handle	ID	Handle
OM2	ObserverModel.pmfEntropy_source_eq_sourceEntropy	PRIV3	AbstractClassSystem.identity_disclosure_separation
	Paper1IT/PMFEntropy.lean		AbstractClassSystem/Extended.lean
ROB1	AbstractClassSystem.OpenWorld.extend_to_force_barrier	ROB2	AbstractClassSystem.OpenWorld.barrierFreedom_not_extension_stable
	AbstractClassSystem/Undecidability.lean		AbstractClassSystem/Undecidability.lean
S1	Ssot.Paper1IT.exists_optimal_budget_split	UND1	AbstractClassSystem.OpenWorld.hasBarrier_not_computable
	Paper1IT/BudgetSplit.lean		AbstractClassSystem/Undecidability.lean
UND2	AbstractClassSystem.OpenWorld.barrierFree_not_computable	ZEC1	Ssot.GraphEntropy.zeroErrorConditionalEntropySandwich
	AbstractClassSystem/Undecidability.lean		Paper1IT/ZeroErrorConditionalEntropy.lean